

PATHWAYS Defense Notes

Our basis for the tool defense. All demo records are made-up (synthetic) seed data, and we should say so at the start.

1. PATHWAYS in one minute

What it is. PATHWAYS is a web-based project information system for humanitarian and development organizations. It keeps project, activity, beneficiary, indicator and budget records in one place. It turns them into dashboards and reports, and it flags problems using simple IF-THEN rules.

What it is not. It is not an AI system and does not predict or forecast. It does not replace KOBO or other collection tools; we import their files (CSV, Excel or PDF). It does not judge individual beneficiaries.

Who uses it. Six roles:

Role	Main job in the system
System Administrator	Manages users, configures alert rules
Project Officer	Encodes and imports field data, submits activity proof
M&E Officer	Builds forms, maps imports, defines indicators, verifies proof and expenses
Project Manager	Creates projects and activities, approves expenses, acts on alerts
Program Manager	Oversees several projects, signs off expenses; sees beneficiary totals only
Grant Manager	Tracks milestones and budget for assigned projects; sees beneficiary totals only

Beneficiaries never log in. Donors and the public only see the approved public tracker.

The flow in one line: set up the project → collect or import data → define indicators → the system computes the numbers → rules raise alerts → a person reviews and decides → generate reports → publish an approved summary.

2. How it is built

- **Frontend:** Next.js and React.
- **Backend:** NestJS with Prisma.
- **Database:** PostgreSQL on Supabase. Login uses Supabase with an authenticator app (two-factor), and files are kept in private storage.
- **Charts and maps:** ECharts and MapLibre.
- **Imports:** SheetJS, PapaParse and unpdf.
- **Testing:** Vitest and Playwright, with automatic checks on GitHub on every change.

Design approach (what we can defend):

- **Metadata-driven.** Forms and fields are configured as data, so new forms do not need new code.
- **Security is checked on the server for every request.** It checks the role, the organization and whether the user is assigned to the project. Hiding a button is not the security.
- **Numbers are computed in the database, not in the browser,** from approved records only.
- **Rules are fixed IF-THEN conditions.** The same data always gives the same result.

If asked "why this stack?": the design principles above are written in our design docs. A detailed comparison against other frameworks was not documented, so present our reasons as our own team decision.

3. Main features

Feature	What it does	Who mainly uses it
Projects and activities	Project profile, activities, proof of completion, overdue reasons, deadline extensions	PM, PO, M&E
Budget and expenses	Planned budget; expenses go submit → verify → approve → sign-off, each by a different person	PO, M&E, PM, PgM/GM
Beneficiary profiles	Registration with consent, duplicate check; viewing details needs a second login check	PO, M&E, PM
Beneficiary journey	Stages, participation and follow-up events; corrections are added, never overwritten	PO, M&E, PM
Forms and imports	Build forms; upload files, match columns to fields, validate, then save	M&E, PO
Indicators	Baseline, target, direction (higher or lower is better), measurements; shared indicator library	M&E, PM
Dashboards and SADDD	Role dashboards; breakdown by sex, age and disability	All roles (totals only for PgM/GM)
Alerts and recommendations	IF-THEN rules raise alerts with ready-made recommendations; a person decides	Admin sets up; all roles review
Reports	Six report types in CSV, Excel or PDF	All roles (by permission)
Public tracker	Approved project summary for donors; must be approved by a second person	Admin, PM, PgM, GM

4. How the main numbers are computed

Number	Simple formula	Notes
Indicator progress	$(\text{actual} - \text{baseline}) \div (\text{target} - \text{baseline}) \times 100$	How far we moved from baseline to target. It is not a pass or fail score. It can go over 100%.
KPI achievement	Average progress of indicators that have data	Indicators with no data are left out, not counted as zero.
Budget utilization	$\text{Approved expenses} \div \text{planned project budget} \times 100$	Only <i>approved</i> expenses count; pending and rejected ones do not.
Efficiency ratio	$\text{KPI achievement} \div \text{budget utilization}$	Above 1 suggests progress is ahead of spending.
Timeline progress	$\text{Days elapsed} \div \text{total project days} \times 100$	Shows how much of the project period has passed.
Activity completion	$\text{Completed activities} \div \text{all activities (except cancelled)} \times 100$	
Overdue activity	Planned end date has passed and it is not completed	Overdue activities need a written explanation.
Milestones on time	$\text{Milestones finished on or before target date} \div \text{finished milestones} \times 100$	
SADDD	Counts of beneficiaries by sex, age group (0-9, 10-14, 15-17, 18-24, 25+, unknown) and disability	Shown only after the project period closes. If any group has 1-4 people, the table is hidden to protect privacy.
Survey improvement	$\text{Average post-test score minus average pre-test score (scores turned into \%)}$	Only shown when there are at least 5 matched pre/post pairs.
Evaluation score	$\text{Sum of (score} \div \text{max score} \times \text{weight) for each criterion; weights add up to 100}$	Used for the project evaluation (OECD-DAC criteria).

Worked example (made-up project):

Indicator	Baseline → target	Actual	Progress
A	10 → 30	20	50%
B	0 → 100	75	75%
C	(no data yet)	–	not counted
D (lower is better)	12 → 4	9	37.5%

- KPI achievement = $(50 + 75 + 37.5) \div 3 = 54.2\%$
- Budget: ₱150,500.50 approved out of ₱500,000 planned = **30.1%**
- Efficiency = $54.2 \div 30.1 = 1.8$, so progress is ahead of spending.

We checked these numbers by running the system's own calculation code.

5. How alerts and recommendations work

1. The System Administrator writes a rule. For example:

IF activities overdue is at least 2 **THEN** raise a **Medium** alert, and recommend: *"Reschedule delayed activities"* and *"Reassign field support."*

2. A rule can check 11 things:

- indicator value
- indicator progress
- timeline elapsed
- days remaining
- days overdue (project)
- activity completion
- number of overdue activities
- days an activity is overdue
- budget utilization
- beneficiaries needing follow-up
- survey improvement

3. **Conditions.** The comparisons are: less than, at most, equal to, at least, more than, and between. Conditions can be combined with AND or OR.

4. **Each project has its own thresholds.** In our demo, one project alerts at 2 or more overdue activities, while another alerts only above 3.

5. **The system checks the rules automatically,** every 5 minutes and every hour, once the scheduler is switched on.

6. **Each alert opens only once, not over and over.** It **closes by itself** when the problem goes away. If data is missing, the rule does **not** fire, so there are no false alarms from empty data.

7. **The recommendations are written in advance by the administrator** and attached to the rule. The system does **not** generate advice.

8. **A person always decides.** They review the alert and record **Accept, Partially accept, Decline or Escalate**, with a note. Every action is logged.

Example checks (made-up data), rule "budget used is at least 90%":

Budget used	Result
95%	Alert raised

Budget used	Result
85%	No alert, and an open alert closes itself
Exactly 90%	Alert raised, because "at least" includes 90
No budget entered	No alert, because missing data never triggers

Good performance (favorable) rules. You *can* write a rule like "progress is at least 100%". But the system still shows it as an alert; there is no separate "good news" type yet. Present this honestly as future work. The current focus is catching problems early so the team can fix the current project. It does not suggest plans for future projects.

6. What we can and cannot say about beneficiaries

What we CAN say (project level):

- How many people were registered and enrolled, and their breakdown by sex, age and disability.
- Who attended which activity, and which journey stages they reached.
- How many need follow-up.
- For a group of at least 5, the average change between pre-test and post-test.

What we CANNOT say:

- That a specific person **learned, improved, succeeded or failed**. Our scope says we do not evaluate beneficiaries as individuals.
- That attending sessions or reaching a stage proves learning. That is an **output**, not an outcome.
- That the project **caused** an improvement. There is no comparison group.

What would be needed for a stronger claim:

- Pre-test and post-test scores recorded per person through the system.
- Outcome indicators measured over time, such as skills, school retention or employment.
- Follow-up records after the project.
- A comparison group, if we want to claim impact.

Safe phrase to use: *"PATHWAYS supports project-level outcome monitoring, not individual evaluation."*

7. Things to be careful about (and how to answer)

Topic	What is true now	How to say it
Budget % on different screens	The Analytics card and the budget rule divide by the total budget plus activity budget lines, so they can show a lower % than the project overview	"Please refer to the project overview figure; aligning the Analytics view is a registered fix."
"Active budget alerts" card	It actually counts all open alerts, not just budget alerts	Avoid this card, or call it "open alerts"
Survey and evaluation scores	There is no screen yet to enter pre/post scores or evaluation scores; the demo values come from seed data	"The calculation is built; the input screen is future work."
Beneficiary reach on the monitoring dashboard	Participation counts are held back for privacy and show "Not available"	"Withheld by design until the privacy release is approved."
Automatic alert checking online	The scheduler is not yet fully switched on in the hosted version	Run the check once before the demo, or say "shown after an evaluation run."
User acceptance testing (ISO/IEC 25010)	Not yet conducted, so Objective 3 has no results yet	"The survey instrument is ready; results will follow the UAT."

Topic	What is true now	How to say it
Reusable project templates (R4)	Only the indicator library is built	"Reusable indicators are done; full project templates are future work."
"Saves about 40 hours"	It is an estimate from staff, not measured	"An estimate, to be measured during UAT."
"Prescriptive analytics"	We only have rule-based alerts with prewritten recommendations	Say "rule-based decision support."

8. Wording to fix in the manuscript

Current wording	Safer wording
"prescriptive analytics proposing courses of action"	"rule-based decision support with predefined recommendations"
"encode, organize, monitor and evaluate beneficiary participation"	"...and monitor beneficiary participation"
"improves beneficiary outcome assessment"	"supports project-level outcome monitoring"
"suggest actions for improvement" (Objective 1.7)	"present predefined recommendations configured by administrators"
"reusable project structures" (R4)	"reusable indicator definitions; project templates as future work"

9. Likely questions and short answers

- 1. What problem does it solve?** Data is scattered across tools and repeatedly exported and cleaned. PATHWAYS keeps it in one place and turns it into dashboards, alerts and reports.
- 2. Is it AI?** No. Every alert comes from a fixed IF-THEN rule that a person configures. The same data always gives the same answer.
- 3. Where do dashboard numbers come from?** They are computed by the database from approved records. Missing data is shown as "not available", not as zero.
- 4. How is KPI achievement computed?** It is the average of each indicator's progress from baseline to target, counting only indicators with data.
- 5. What does 100% mean?** The target was reached. It is not a grade.
- 6. How does data import work?** Upload the file. The system suggests which column goes to which field and the M&E Officer can adjust it. Each row is checked, and invalid rows are held back, not saved.
- 7. Who can change alert thresholds?** Only the System Administrator. Each project can have its own threshold, and every change is logged.
- 8. Does the system give advice on its own?** No. Recommendations are written in advance and a person decides what to do with them.
- 9. What if data is missing or updated later?** Missing data never triggers an alert. When data is updated, the rule is checked again, and the alert closes by itself if the problem is gone.
- 10. Can it flag good performance?** A rule can be written for it, but it is still shown as an alert. A positive alert type is future work.
- 11. Can you tell if a beneficiary improved?** Not individually. We only report group averages, for 5 or more people.
- 12. How is beneficiary privacy protected?** Through role and project checks on every request, and a second login check to view personal details. Program and Grant Managers see totals only. Groups smaller than 5 are hidden, and nothing personal goes to the public tracker.
- 13. Does it replace Kobo?** No. It imports files from tools like Kobo.

14. **How was it tested?** Through over 1,000 automated tests, which all passed in our latest run, plus manual checks of the calculations. User acceptance testing is still to be done.
15. **What is not finished?** The input screens for survey and evaluation scores, project templates, a positive alert type, full online scheduling, and the UAT.

10. Quick demo flow (about 2 minutes)

Before starting: reload the demo data, run the alert check once, and log in with the authenticator. Say: "All records here are synthetic."

1. **Dashboard** (Project Manager): project list with KPI, budget and overdue counts.
2. **Activities**: an overdue activity and its required explanation.
3. **Budget**: planned versus approved spending, and the verify → approve → sign-off steps.
4. **Alerts**: open an alert. It shows the metric, the actual value, the threshold and the prewritten recommendation.
5. **Record a decision**: choose Accept and add a note. The alert moves to "Actioned".
6. **Analytics** (M&E Officer): SADDD breakdown for the closed project.
7. **Reports**: generate a project summary PDF.
8. **Public tracker**: only the approved summary is visible.

Where to point if the panel asks for proof

- Alert rules and checks: `apps/api/src/modules/rules/` and migration `0060_rules_budget_beneficiary_survey_metrics`
- Calculations: `packages/shared/src/monitoring/` (progress, KPI, budget, timeline)
- Roles and permissions: `apps/api/src/modules/auth/rbac-contract.json`
- SADDD and privacy: migration `0028_revised_aggregate_permission_guards`
- Seed rules used in the demo: `apps/api/prisma/local-demo-data.ts`
- Known gaps list: `docs/deferred-features.md`